

E023

Product Design

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A team has one week to learn whether warehouse pickers understand a new three-step order flow. The design lead has a full license for a professional design tool, and the company recently updated its brand colors. What kind of prototype should they build?

- A. The fully coded production feature.
- B. A polished, high-fidelity prototype using the new brand colors.
- C. A low-fidelity wireframe or paper prototype of the three steps.
- D. No prototype; ask pickers to describe their ideal flow.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

The flow has been validated in two rounds. The team now needs feedback on animation timing and visual hierarchy before handoff. What fidelity fits?

- A. A written specification only.
- B. Low fidelity, since early prototypes are always best.
- C. Paper prototype.
- D. High fidelity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A startup with a \$2,000 research budget wants a quick usability test of a prototype for booking pet grooming. Its founder previously worked at a large bank, and the app will launch in two cities. How many participants should the first round include?

- A. Every future customer, since any unseen user may find a problem.
- B. About five.
- C. At least 200, for statistical validity.
- D. One per city, so two.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

After testing with five users and fixing the issues found, what is the best next step?

- A.** Switch to a survey of preferences.
- B.** Launch, since five users found all the problems.
- C.** Run another small round on the revised design.
- D.** Retest the original design with 50 users.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

During a moderated test in a bright conference room, a participant pauses on the checkout screen. The moderator, who wrote the prototype's code, says, "The button you want is on the right—do you like that layout?" What is wrong with this moderation?

- A.** Nothing; helping keeps the session moving.
- B.** It gives the answer and asks for opinion instead of observing what the user does.
- C.** The room was too bright.
- D.** The moderator should have explained the whole design before the task.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In concept tests, users loved a feature that translates live video calls into sign language with a 3D avatar. Engineers say reliable real-time sign translation is not possible with current tools. The concept board was printed on glossy paper. What is the best assessment?

- A.** Desirable but not currently feasible; rescope toward what can be built reliably.
- B.** Proceed, because user love outweighs other factors.
- C.** Drop all accessibility features, since this one is hard.
- D.** It fails viability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

An expense app lets users submit a report with no receipts attached, and only later emails them that it was rejected. The app has 30,000 users and a dark mode. Which heuristic does the design most clearly violate?

- A. Recognition rather than recall.
- B. Aesthetic and minimalist design.
- C. Help users recover from errors.
- D. Error prevention.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A file upload shows no progress bar or message, so users click “upload” several times. Which heuristic is violated?

- A. Match between system and real world.
- B. Error prevention.
- C. Consistency and standards.
- D. Visibility of system status.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A product team at a 60-person clinic software company is writing backlog items for a refill feature. Its sprint length is two weeks. Which is the best-formed user story?

- A. “As a patient, I want to request a refill online so that I don’t have to call the clinic.”
- B. “Build refills in two weeks.”
- C. “The system shall store refill requests in a table.”
- D. “As a patient, I want to request a refill online.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Twelve participants tried to complete checkout on a prototype built over a long weekend. Nine finished without help, two finished only after the moderator helped, and one gave up. What was the unassisted completion rate?

- A. About 82%
- B. 75%
- C. 25%
- D. About 92%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A team’s target is an 80% unassisted completion rate with 15 participants. How many must finish without help to meet it?

- A. 8
- B. 3
- C. 12
- D. 15, since any failure means the design is broken.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A banking dashboard marks overdue bills only by turning their text red. The design team uses a popular component library, and the dashboard is used by 50,000 customers. What is the main accessibility problem?

- A. Color alone conveys status; add text or an icon such as “Overdue.”
- B. Accessibility only matters for blind users, who use screen readers anyway.
- C. There is no problem because the library is widely used.
- D. Red is not a brand color.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A team has one week to learn whether warehouse pickers understand a new three-step order flow. ~~The design lead has a full license for a professional design tool, and the company recently updated its brand colors.~~ What kind of prototype should they build?

Target word: fidelity for an early flow test **Decides:** The question is about the flow, early, with little time.

A	Adjacent category	Too costly to change if the flow is wrong.
B	Camouflage	Brand colors and tools do not decide; polish is not what is being tested.
C	Correct	Tests the flow fast and cheaply.
D	Dropped element	Drops the artifact needed to observe behavior.

2. Answer D Concept 1 • transfer

The flow has been validated in two rounds. The team now needs feedback on animation timing and visual hierarchy before handoff. What fidelity fits?

Target word: reverse: late-stage detail **Decides:** Visual and interaction detail needs high fidelity.

A	Adjacent category	No way to observe people using it.
B	Absolute language	Fidelity should match the question.
C	Inverted operation	Cannot show animation or real visual hierarchy.
D	Correct	Timing and visual detail can only be judged in a polished prototype.

3. Answer B Concept 2

A startup with a \$2,000 research budget wants a quick usability test of a prototype for booking pet grooming. ~~Its founder previously worked at a large bank, and the app will launch in two cities.~~ How many participants should the first round include?

Target word: sample size for a usability round **Decides:** About five users find most major issues.

A	Absolute language	Returns fall quickly after about five users per round.
B	Correct	Five users typically surface most major usability problems; then iterate.
C	Adjacent category	That is survey logic; usability tests find problems, not percentages.
D	Camouflage	The launch cities do not decide sample size.

4. Answer C Concept 2 • transfer

After testing with five users and fixing the issues found, what is the best next step?

Target word: reverse: what follows a small round **Decides:** Small tests work in iterations.

A	Confusion set	Preferences are not usability evidence.
B	Absolute language	Five users find most, not all, problems.
C	Correct	Iterative rounds catch new and remaining problems.
D	Inverted operation	Tests the old design, not the fixes.

5. Answer B Concept 3

During a moderated test in a bright conference room, a participant pauses on the checkout screen. The moderator, who wrote the prototype's code, says, "The button you want is on the right—do you like that layout?" What is wrong with this moderation?

Target word: moderation error **Decides:** The moderator led the participant and asked for opinion.

A	Inverted operation	Helping hides the very problem being tested.
B	Correct	Leading and opinion questions spoil behavioral evidence.
C	Camouflage	The room does not decide the problem.
D	Adjacent category	Pre-explaining also biases behavior.

6. Answer A Concept 4

In concept tests, users loved a feature that translates live video calls into sign language with a 3D avatar. Engineers say reliable real-time sign translation is not possible with current tools. The concept board was printed on glossy paper. What is the best assessment?

Target word: which lens fails **Decides:** Desirable but not feasible.

A	Correct	All three lenses must pass.
B	Dropped element	Drops feasibility.
C	Absolute language	One infeasible feature does not rule out others.
D	Confusion set	The stated problem is technical, not business.

7. Answer D Concept 5

An expense app lets users submit a report with no receipts attached, and only later emails them that it was rejected. The app has 30,000 users and a dark mode. Which heuristic does the design most clearly violate?

Target word: name the heuristic **Decides:** The error could be blocked before it happens.

A	Adjacent category	About memory load, not blocking mistakes.
B	Camouflage	Dark mode and looks are not the issue.
C	Confusion set	That covers error messages after the fact; the bigger miss is allowing the error.
D	Correct	The app should stop incomplete submissions up front.

8. Answer D Concept 5 • transfer

A file upload shows no progress bar or message, so users click "upload" several times. Which heuristic is violated?

Target word: heuristic from a symptom **Decides:** Users cannot see what the system is doing.

A	Inverted operation	About familiar language, not status.
B	Adjacent category	Related, but the root is missing feedback.
C	Confusion set	About matching conventions, not feedback.
D	Correct	The system should show that the upload is in progress.

9. Answer A Concept 6

A product team at a ~~60-person clinic software company~~ is writing backlog items for a refill feature. ~~Its sprint length is two weeks.~~ Which is the best-formed user story?

Target word: well-formed story **Decides:** Who, what, and why.

A	Correct	Names the user, the goal, and the benefit.
B	Camouflage	Echoes the sprint length; no user or value.
C	Adjacent category	A system requirement, not a user story.
D	Dropped element	Missing the “so that” benefit.

10. Answer B Concept 7

Twelve participants tried to complete checkout on a prototype ~~built over a long weekend~~. Nine finished without help, two finished only after the moderator helped, and one gave up. What was the unassisted completion rate?

Target word: unassisted completion rate **Decides:** 9 of 12 finished without help.

A	Dropped element	Drops the one who gave up: $9 \div 11$.
B	Correct	$9 \div 12 = 75\%$.
C	Inverted operation	That is the rate of people who did not finish unassisted.
D	Adjacent category	Counts assisted completions as successes: $11 \div 12$.

11. Answer C Concept 7 · transfer

A team’s target is an 80% unassisted completion rate with 15 participants. How many must finish without help to meet it?

Target word: solve for successes **Decides:** 15×0.80 .

A	Wrong stem number	Reads 80% as 8 people.
B	Inverted operation	That is how many may fail.
C	Correct	$15 \times 0.8 = 12$.
D	Absolute language	The target allows some failures.

12. Answer A Concept 8

A banking dashboard marks overdue bills only by turning their text red. ~~The design team uses a popular component library, and the dashboard is used by 50,000 customers.~~ What is the main accessibility problem?

Target word: accessibility issue **Decides:** Color is the only signal.

A	Correct	Users with color-vision differences may miss red-only cues.
B	Absolute language	Accessibility covers many needs, including color vision.
C	Camouflage	A popular library does not guarantee accessible use.
D	Adjacent category	Brand fit is not accessibility.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Prototype fidelity	Low-fidelity prototypes (sketches, wireframes, paper) test concepts and flows early and cheaply; high fidelity tests visual detail and interaction polish later.	Building a polished prototype before the flow is validated.	An early-stage question about structure or flow.
2	Small usability tests	About five participants per round reveal most major usability problems; test, fix, and test again.	Believing a usability test needs hundreds of users to be valid.	A “very limited” test or a small budget.
3	Observe behavior, not opinions	Usability testing watches what people do on realistic tasks (think-aloud) without leading them; what users say they like is weak evidence.	Asking “Do you like it?” or helping a stuck participant.	A moderator question or intervention during a test.
4	Desirability, feasibility, viability	A good design is something users want (desirable), the team can build (feasible), and the business can sustain (viable).	Judging a design only on user appeal.	A concept users love that has a technical or cost problem.
5	Usability heuristics	Key heuristics include visibility of system status, match with the real world, error prevention, consistency, and recognition over recall.	Mixing up error prevention with helpful error messages.	A described interface problem to be named.
6	User stories and acceptance criteria	A user story states who, what, and why (“As a..., I want..., so that...”); acceptance criteria make it testable.	Writing a story about the system or with no stated benefit.	A requirement written in story form.
7	Task completion rate	Unassisted completion rate = participants who finished without help ÷ all participants who attempted.	Counting assisted completions as successes or dropping those who quit.	Counts of completed, assisted, and abandoned attempts.
8	Accessibility basics	Do not rely on color alone; normal text needs at least 4.5:1 contrast (WCAG AA); interfaces should work with keyboard and screen readers.	Treating accessibility as a later add-on or as only for blind users.	Color used as the only signal, or low contrast.

Course scope (public description): Product Design teaches students how to design a technical solution to a business problem, create a low-fidelity prototype of the technical solution, and then conduct a very limited user test using a created prototype. The course covers skills in functional knowledge contribution, product design, user needs analysis, and software feasibility and usability studies. The following course is a prerequisite and should be completed prior to beginning Product Design: Product Strategy.